Meperidine (pethidine): Drug information - UpToDate

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Contributor Disclosures

For additional information see "Meperidine (pethidine): Patient drug information" and "Meperidine (pethidine): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates   August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term

ALERT: US Boxed Warning

Risk of medication errors (oral solution):

Ensure accuracy when prescribing, dispensing, and administering meperidine oral solution. Dosing errors due to confusion between mg and mL, and other meperidine oral solutions of different concentrations, can result in accidental overdose and death.

Addiction, abuse, and misuse:

Because the use of meperidine exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing and reassess all patients regularly for the development of these behaviors and conditions.

Opioid analgesic risk evaluation and mitigation strategy (REMS):

Health care providers are strongly encouraged to complete a REMS-compliant education program and to counsel patients and caregivers on serious risks, safe use, and the importance of reading the Medication Guide with each prescription.

Life-threatening respiratory depression:

Serious, life-threatening, or fatal respiratory depression may occur with use of meperidine, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of meperidine are essential.

Accidental ingestion:

Accidental ingestion of meperidine, especially by children, can result in a fatal overdose of meperidine.

Neonatal opioid withdrawal syndrome:

Advise pregnant women using opioids for an extended period of time of the risk of neonatal opioid withdrawal syndrome (NOWS), which may be life threatening if not recognized and treated. Ensure that management by neonatology experts will be available at delivery.

Cytochrome P450 3A4 interaction:

The concomitant use of meperidine with all cytochrome P450 3A4 inhibitors may result in an increase in meperidine plasma concentrations, which could increase or prolong adverse reactions and may cause potentially fatal respiratory depression. In addition, discontinuation of a concomitantly used cytochrome P450 3A4 inducer may result in an increase in meperidine plasma concentration. Monitor patients receiving meperidine and any CYP3A4 inhibitor or inducer.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of meperidine and benzodiazepines or other CNS depressants for use in patients for whom alternative treatment options are inadequate.

Concomitant use of meperidine with monoamine oxidase inhibitors:

The concomitant use of meperidine with monoamine oxidase inhibitors (MAOIs) can result in coma, severe respiratory depression, cyanosis, and hypotension. Use of meperidine with MAOIs within the last 14 days is contraindicated.

Brand Names: US

- Demerol

Pharmacologic Category

- Analgesic, Opioid

Dosing: Adult

Dosage guidance:

Safety: Consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmefene) for patients with factors associated with an increased risk for overdose, such as history of overdose or substance use disorder, patients with sleep-disordered breathing, higher opioid dosages (≥50 morphine milligram equivalents/day orally), and/or concomitant benzodiazepine use (Ref).

Acute pain

Acute pain (alternative therapy):

Note: Other than in rare situations, **NOT** recommended for the treatment of pain due to potential neurotoxicity and availability of safer alternatives, especially in patients with kidney disease or elderly patients. Oral route and patient-controlled analgesia (PCA) use are **NOT** recommended (Ref). Reserve for patients who do not tolerate or have no access to other options.

IM (preferred route), IV: 50 to 150 mg every 3 to 4 hours as needed; maximum daily dose: 600 mg; limit duration to ≤48 hours (Ref). If IV administration is required, administer diluted and at a slow rate. Dosing based on severity of pain; start at the lower end of dosing range.

Obstetrical analgesia: **IM, SUBQ:** 50 to 100 mg when pain becomes regular; may repeat at 1- to 3-hour intervals.

Preoperative: **IM, SUBQ:** 50 to 100 mg administered 30 to 90 minutes before the beginning of anesthesia.

Postoperative shivering

Postoperative shivering (off-label use): IV: 12.5 to 50 mg once (Ref) **or** 0.2 mg/kg with adjunctive dexamethasone (Ref).

Rigors from amphotericin B

Rigors from amphotericin B (conventional) (off-label use): IV: 25 to 50 mg once (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Single doses for the treatment of shivering/rigors may be considered in patients with reduced kidney function or on renal replacement therapy; use with caution (Ref).

Altered kidney function:

eGFR ≥60 mL/minute/1.73 m^2: No dosage adjustment necessary (Ref).

eGFR <60 mL/minute/1.73 m^2: Avoid use; normeperidine (active metabolite) may accumulate and increase risk of neurotoxicity including delirium, myoclonus, altered mental status, and seizures (Ref).

Augmented renal clearance (measured urinary CrCl ≥130 mL/minute/1.73 m^2):

Note: Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Younger patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients (Ref).

No dosage adjustment necessary; however, may require increased doses due to increased renal clearance (Ref).

Hemodialysis, intermittent (thrice weekly): Avoid use (Ref).

Peritoneal dialysis: Avoid use (Ref).

CRRT: Avoid use (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Avoid use (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution and titrate slowly; monitor closely for signs of CNS excitation (eg, seizure activity) and CNS and respiratory depression. In patients with severe impairment, consider a lower dose when initiating therapy; an increased opioid effect may be seen in patients with cirrhosis; dose reduction is more important for the oral (route not recommended (Ref) than IV route (Ref).

Dosing: Older Adult

Avoid use as an analgesic (Ref).

Dosing: Pediatric

(For additional information see "Meperidine (pethidine): Pediatric drug information")

Acute pain

Acute pain (analgesic): Limited data available:

Note: Although FDA approved, the American Pain Society (2016) and ISMP (2007) do not recommend meperidine use as an analgesic. If use for acute pain (in patients without renal or CNS disease) cannot be avoided, treatment should be limited to ≤48 hours and doses should not exceed 600 mg per 24 hours in adults. Oral route is not recommended for treatment of acute or chronic pain. If IV route is required, consider a reduced dose. Patients with prior opioid exposure may require higher initial doses. Should not be used for chronic pain. Doses should be titrated to appropriate analgesic effect; when changing route of administration, note that oral doses are about half as effective as parenteral dose.

Infants >6 months, Children, and Adolescents:

IM, IV, or SUBQ: Initial: 0.8 to 2 mg/kg/dose every 3 to 4 hours as needed; maximum dose range: 50 to 75 mg/dose; consider dose reduction in critically ill (Ref).

Oral: Initial: 1.1 to 3 mg/kg/dose every 3 to 4 hours as needed; maximum dose range: 50 to 100 mg/dose (Ref).

Sedation, preoperative

Sedation, preoperative: Limited data available: Infants, Children, and Adolescents: IM, IV, SUBQ: 0.5 to 2 mg/kg administered 30 to 90 minutes before the beginning of anesthesia; maximum dose: 2 mg/kg or 100 mg/dose, whichever is less (Ref).

Sickle cell disease, acute crisis

Sickle cell disease, acute crisis: Limited data available: **Note:** Due to risk of adverse effects from metabolite (normeperidine) accumulation, meperidine is unlikely first-line agent and generally not recommended for use unless it is the only opioid effective for the patient or the patient has uncorrectable intolerances or allergies to other opioid options (Ref).

Infants ≥6 months, Children, and Adolescents:

Patient weight <50 kg: IV: Initial: 0.75 to 1 mg/kg every 3 to 4 hours as needed (Ref).

Patient weight ≥50 kg: IV: Initial: 50 to 150 mg every 3 hours as needed (Ref).

Shivering, postoperative

Shivering, postoperative: Limited data available: Infants, Children, and Adolescents: IV: 1 to 2 mg/kg/dose once; maximum dose range: 50 to 75 mg/dose; consider dose reduction in critically ill (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Avoid use as an analgesic in renal impairment (Ref).

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution and titrate slowly; monitor closely for signs of CNS excitation (eg, seizure activity) and CNS and respiratory depression. In patients with severe impairment, consider a lower dose when initiating therapy; an increased opioid effect may be seen in patients with cirrhosis; dose reduction is more important for the oral route (due to increased bioavailability) than IV route based on experience in adults (Ref).

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Bradycardia, circulatory depression, facial flushing, hypotension, palpitations, shock, syncope, tachycardia

Dermatologic: Diaphoresis, pruritus, skin rash, urticaria

Endocrine & metabolic: Antidiuretic effect

Gastrointestinal: Biliary colic, constipation, nausea, vomiting, xerostomia

Genitourinary: Urinary retention

Hypersensitivity: Histamine release, hypersensitivity reaction (including anaphylaxis)

Local: Injection-site reaction (including induration at injection site, injection-site phlebitis, irritation at injection site, pain at injection site, postinjection flare, urticaria at injection site)

Nervous system: Agitation, asthenia, confusion, delirium, disorientation, dizziness, drug abuse, hallucination, headache, involuntary muscle movements (including muscle twitching, myoclonus), mood changes (including euphoria, dysphoria), neonatal withdrawal, opioid dependence, sedated state, seizure, tremor

Ophthalmic: Visual disturbance

Respiratory: Respiratory depression

Postmarketing:

Genitourinary: Hypogonadism (Brennan 2013; Debono 2011)

Nervous system: Allodynia (opioid-induced hyperalgesia) (FDA Safety Communication 2023)

Contraindications

Hypersensitivity (eg, anaphylaxis) to meperidine or any component of the formulation; use with or within 14 days of MAO inhibitors; significant respiratory depression; acute or severe bronchial asthma in an unmonitored setting or in the absence of resuscitative equipment; GI obstruction, including paralytic ileus (known or suspected).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in US labeling): Known or suspected mechanical GI obstruction (eg, bowel obstruction or strictures) or any diseases/conditions that affect bowel transit (eg, ileus of any type); suspected surgical abdomen (eg, acute appendicitis or pancreatitis); mild pain that can be managed with other pain medications; acute or severe bronchial asthma, chronic obstructive airway, status asthmaticus; acute respiratory depression; hypoxia; hypercapnia; cor pulmonale; acute alcoholism, delirium tremens, and convulsive disorders; severe CNS depression, increased cerebrospinal or intracranial pressure and head injury; concurrent use or use within 14 days of an MAOI.

Warnings/Precautions

Concerns related to adverse effects:

• CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving).

• CNS events: Normeperidine (an active metabolite and CNS stimulant) may accumulate and precipitate anxiety, tremors, or seizures; risk increases with preexisting CNS or renal dysfunction, prolonged use (>48 hours), and cumulative dose (>600 mg/24 hours in adults). Oral meperidine should not be used since first-pass metabolism decreases efficacy while increasing normeperidine concentrations (APS 2016). **Note:** Naloxone does not reverse, and may even worsen, neurotoxicity.

• Constipation: May cause constipation which may be problematic in patients with unstable angina and patients post-myocardial infarction (MI). Consider preventive measures (eg, stimulant laxative) to reduce the potential for constipation.

• Esophageal dysfunction: Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.

• Hyperalgesia: Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.

• Hypotension: May cause severe hypotension (including orthostatic hypotension and syncope); use with caution in patients with hypovolemia, cardiovascular disease (including acute MI), or drugs which may exaggerate hypotensive effects (including phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration. Avoid use in patients with circulatory shock.

• Respiratory depression: Fatal respiratory depression may occur. Carbon dioxide retention from opioid-induced respiratory depression can exacerbate the sedating effects of opioids. Patients and caregivers should be educated on how to recognize respiratory depression and the importance of getting emergency assistance immediately (eg, calling 911) in the event of known or suspected overdose.

• Serotonin syndrome: May occur with concomitant use of serotonergic agents (eg, selective serotonin reuptake inhibitors, serotonin-norepinephrine reuptake inhibitors, triptans, tricyclic antidepressants), lithium, St. John's wort, agents that impair metabolism of serotonin (eg, monoamine oxidase inhibitors [MAOIs]). Monitor patients for serotonin syndrome such as mental status changes (eg, agitation, hallucinations, coma); autonomic instability (eg, tachycardia, labile blood pressure, hyperthermia); neuromuscular changes (eg, hyperreflexia, incoordination); and/or GI symptoms (eg, nausea, vomiting, diarrhea).

Disease-related concerns:

• Abdominal conditions: May obscure diagnosis or clinical course of patients with acute abdominal conditions.

• Adrenocortical insufficiency: Use with caution and reduce initial dosage in patients with adrenal insufficiency, including Addison disease. Long-term opioid use may cause secondary hypogonadism, which may lead to mood disorders and osteoporosis (Brennan 2013).

• Biliary tract impairment: Use with caution in patients with biliary tract dysfunction, including acute pancreatitis; opioids may cause constriction of sphincter of Oddi.

• CNS depression/coma: Avoid use in patients with impaired consciousness or coma as these patients are susceptible to intracranial effects of carbon dioxide retention.

• Delirium tremens: Use with caution in patients with delirium tremens.

• Head trauma: Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur.

• Hepatic impairment: Use with caution in patients with hepatic disorders; meperidine and to a lesser degree normeperidine may accumulate and precipitate either CNS depression or CNS excitation (eg, anxiety, tremors, or seizures) (Danzinger 1994; Tegeder 1999).

• Obesity: Use with caution in patients who are morbidly obese.

• Pheochromocytoma: Use with caution in patients with pheochromocytoma.

• Prostatic hyperplasia/urinary stricture: Use with caution and reduce initial dosage in patients with prostatic hyperplasia and/or urinary stricture.

• Psychosis: Use with caution in patients with toxic psychosis.

• Renal impairment: Avoid use in patients with renal impairment; normeperidine (active metabolite) may accumulate and increase risk of neurotoxicity including delirium, myoclonus, altered mental status, and seizures (ISMP 2007; Kaiko 1983; Koncicki 2017; Owsiany 2019; expert opinion).

• Respiratory disease: Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale, and those with a substantially decreased respiratory reserve, hypoxia, hypercapnia, or preexisting respiratory depression, particularly when initiating and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages. Consider the use of alternative nonopioid analgesics in these patients.

• Seizure disorders: Use with caution in patients with seizure disorders, may cause or aggravate seizures if high doses used or from prolonged use (accumulation of metabolite).

• Sickle-cell disease: In patients with sickle cell disease, use with caution; normeperidine (active metabolite) may accumulate and induce seizures in these patients. Meperidine is not recommended for use in sickle cell patients by the American Pain Society and should only be used in sickle cell patients with a vaso-occlusive crisis if it is the only effective opioid for an individual patient, as normeperidine (active metabolite) may accumulate and induce seizures (APS 2016; NHLBI 2014).

• Sleep-related disorders: Use with caution in patients with sleep-related disorders, including sleep apnea, due to increased risk for respiratory and CNS depression. Monitor carefully and titrate dosage cautiously in patients with mild sleep-disordered breathing. Avoid opioids in patients with moderate to severe sleep-disordered breathing (CDC [Dowell 2022]).

• Tachycardia: Use with caution in patients with atrial flutter and other supraventricular tachycardias; use may increase ventricular response rate possibly due to a vagolytic effect.

• Thyroid dysfunction: Use with caution in patients with thyroid dysfunction, including hypothyroidism.

Concurrent drug therapy issues:

• Benzodiazepines or other CNS depressants: Concomitant use may result in respiratory depression and sedation, which may be fatal. Consider prescribing naloxone or nalmefene for emergency treatment of opioid overdose in patients taking benzodiazepines or other CNS depressants concomitantly with opioids.

Special populations:

• Cachectic or debilitated patients: Use with caution in cachectic or debilitated patients; there is a greater potential for critical respiratory depression, even at therapeutic dosages; reduce initial dosage. Consider the use of alternative nonopioid analgesics in these patients.

• Older adult: Use opioids with caution in older adults; may be more sensitive to adverse effects. Clearance may also be reduced in older adults (with or without renal impairment), resulting in a narrow therapeutic window and increased adverse effects. Monitor closely for adverse effects associated with opioid therapy (eg, respiratory and CNS depression, falls, cognitive impairment, constipation) (CDC [Dowell 2022]). Consider the use of alternative nonopioid analgesics in these patients when possible. Avoid the use of meperidine for pain control, especially in older adults and renally compromised patients because of the risk of neurotoxicity (ISMP 2007). Meperidine should be avoided in those older adults with, or at risk for, delirium because of the potential to cause or worsen delirium.

• Neonates: Neonatal withdrawal syndrome: Signs and symptoms include irritability, hyperactivity and abnormal sleep pattern, high pitched cry, tremor, vomiting, diarrhea, and failure to gain weight. Onset, duration, and severity depend on the drug used, duration of use, maternal dose, and rate of drug elimination by the newborn.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.

• Parenteral: Administer IV injections very slowly, preferably in the form of a diluted solution. Do not administer IV unless a opioid antagonist and the facilities for assisted or controlled respiration are immediately available. When meperidine is given parenterally, especially IV, the patient should be lying down.

• Sulfites: Some preparations may contain sulfites which may cause allergic reaction.

Other warnings/precautions:

• Abrupt discontinuation/withdrawal: Abrupt discontinuation in patients who are physically dependent on opioids has been associated with serious withdrawal symptoms, uncontrolled pain, attempts to find other opioids (including illicit), and suicide. Use a collaborative, patient-specific taper schedule that minimizes the risk of withdrawal, considering factors such as current opioid dose, duration of use, type of pain, and physical and psychological factors. Monitor pain control, withdrawal symptoms, mood changes, suicidal ideation, and for use of other substances and provide care as needed. Concurrent use of mixed agonist/antagonist analgesics (eg, pentazocine, nalbuphine, butorphanol) or partial agonist (eg, buprenorphine) analgesics may also precipitate withdrawal symptoms and/or reduced analgesic efficacy in patients following prolonged therapy with mu opioid agonists.

• Abuse/misuse/diversion: Use with caution in patients with a history of substance use disorder; potential for drug dependency exists. Other factors associated with increased risk for misuse include concomitant depression or other mental health conditions, higher opioid dosages, long-term use, or taking other CNS depressants. Consider offering an opioid overdose reversal agent (eg, naloxone, nalmefene) prescription in patients with an increased risk for overdose, such as history of overdose or substance use disorder, higher opioid dosages (≥50 morphine milligram equivalents [MME]/day orally), concomitant benzodiazepine use, and patients at risk for returning to a high dose after losing tolerance (CDC [Dowell 2022]).

• Acute and/or cancer pain management: Meperidine offers no advantage over other opioids as an analgesic and has unique neurotoxicity. The use of meperidine in this setting should be avoided (APS 2016; ISMP 2007).

• Chronic pain management: Use is not recommended for the management of chronic pain.

• Opioid overdose reversal agent access: Discuss the availability of opioid overdose reversal agents (eg, naloxone, nalmefene) with all patients who are prescribed opioid analgesics, as well as their caregivers, and consider prescribing it to patients who are at increased risk of opioid overdose. These include patients who are also taking benzodiazepines or other CNS depressants, have an opioid use disorder (OUD) (current or history of), or have experienced opioid-induced respiratory depression/opioid overdose. Additionally, health care providers should consider prescribing an opioid overdose reversal agent to patients prescribed medications to treat OUD; patients at risk of opioid overdose even if they are not taking an opioid analgesic or medication to treat OUD; and patients taking opioids, including methadone or buprenorphine for OUD, if they have household members, including children, or other close contacts at risk for accidental ingestion or opioid overdose. Inform patients and caregivers on options for obtaining opioid overdose reversal agents (eg, by prescription, OTC, directly from a pharmacist, a community-based program) as permitted by state dispensing and prescribing guidelines. Educate patients and caregivers on how to recognize respiratory depression, proper administration and important differences of opioid overdose reversal agents, and getting emergency help even if an opioid overdose reversal agent is administered.

• Optimal regimen: An opioid-containing analgesic regimen should be tailored to each patient's needs and based upon the route of administration, degree of tolerance for opioids (naive versus chronic user), age, weight, and medical condition. The optimal analgesic dose varies widely among patients; doses should be titrated to pain relief/prevention.

• Surgery: Opioids decrease bowel motility; monitor for decreased bowel motility in postoperative patients receiving opioids. Use with caution in the perioperative setting; individualize treatment when transitioning from parenteral to oral analgesics.

Warnings: Additional Pediatric Considerations

Due to decreased elimination rate, neonates and young infants may be at higher risk for adverse effects, especially respiratory depression; use with extreme caution and in reduced doses in this age group.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection, as hydrochloride:

Demerol: 25 mg/mL (1 mL [DSC]); 50 mg/mL (1 mL [DSC])

Demerol: 50 mg/mL (30 mL) [contains metacresol]

Solution, Injection, as hydrochloride [preservative free]:

Demerol: 25 mg/mL (1 mL); 50 mg/mL (1 mL); 75 mg/mL (1 mL); 100 mg/mL (1 mL [DSC])

Generic: 25 mg/mL (1 mL); 50 mg/mL (1 mL); 100 mg/mL (1 mL)

Solution, Oral, as hydrochloride:

Generic: 50 mg/5 mL (500 mL)

Tablet, Oral, as hydrochloride:

Generic: 50 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Demerol Injection)

25 mg/mL (per mL): $26.40

50 mg/mL (per mL): $33.60

75 mg/mL (per mL): $7.68

Solution (Meperidine HCl Injection)

25 mg/mL (per mL): $3.04 - $3.05

50 mg/mL (per mL): $3.17

100 mg/mL (per mL): $3.47 - $3.48

Solution (Meperidine HCl Oral)

50 mg/5 mL (per mL): $0.32

Tablets (Meperidine HCl Oral)

50 mg (per each): $44.72 - $47.39

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Generic: 50 mg/mL (1 mL)

Controlled Substance

C-II

Administration: Adult

Injection: Administer IM, SubQ, or IV (patient should be lying down during administration); IV push should be administered slowly using a diluted solution, use of a 10 mg/mL concentration has been recommended. IM administration is preferred when repeated doses are required.

Oral solution: Administer each dose in $^{1}/_{2}$ glass of water (undiluted solution may exert topical anesthetic effect on mucous membranes). Use a calibrated measuring device to measure dosage; do not use a teaspoon or a tablespoon. Use extreme caution; dosing errors can result in accidental overdose and death.

Administration: Pediatric

Oral: Oral solution: Administer each dose in $^{1}/_{2}$ glass of water; undiluted solution may cause topical anesthetic effect on mucous membranes. Use a calibrated measuring device to measure dosage; do not use a teaspoon or a tablespoon. Use extreme caution; dosing errors can result in accidental overdose and death.

Parenteral:

IM: Preferred route when repeat doses required; inject undiluted into a large muscle.

IV: Administer slowly over at least 4 to 5 minutes as a diluted solution (≤10 mg/mL); patients should be lying down; do not administer rapid IV (Dobbins 2010; Gahart 2020; manufacturer's labeling).

SUBQ: Administer undiluted.

Preparation for Administration: Pediatric

Oral: Oral solution: Dilute dose in $^{1}/_{2}$ glass of water prior to use.

Parenteral: IV: Dilute with a compatible solution to ≤10 mg/mL.

Storage/Stability

Injection: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Discard any unused portion of single-dose products after use and multi-dose products after 28 days.

Oral solution, tablets: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Acute pain: Management of acute pain severe enough to require an opioid analgesic and for which alternative treatments are inadequate; obstetrical analgesia, preoperative medication (IV only).

Limitations of use: The American Pain Society and Institute for Safe Medication Practices do not recommend meperidine's use as an analgesic. If use in acute pain (in patients without renal or CNS disease) cannot be avoided, treatment should be limited to ≤48 hours and doses should not exceed 600 mg per 24 hours (APS 2016; ISMP 2007).

Use: Off-Label: Adult

Postoperative shivering; Rigors from amphotericin B (conventional); Targeted temperature management-related shivering

Medication Safety Issues

Sound-alike/look-alike issues:

Meperidine may be confused with meprobamate

Demerol may be confused with Demulen, Desyrel, Dilaudid, Pamelor

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (opioids, all formulations and routes of administration) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Beers Criteria: Meperidine is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to a potentially higher risk of neurotoxicity, including delirium, compared with other opioids; safer alternatives are available. In addition, oral meperidine lacks analgesic efficacy in dosages commonly used (Beers Criteria [AGS 2023]).

Pediatric patients: High-risk medication:

KIDs List: Meperidine is identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list due to risk of acute neurotoxicity (agitation, myoclonus, hyperreflexia, tremors, delirium, and seizures). Use should be avoided in neonates and used with caution in infants, children, and adolescents ≤18 years of age (strong recommendation; high quality of evidence) (PPA [McPherson 2025]).

Other safety concerns:

Avoid the use of meperidine for pain control, especially in elderly and renally compromised patients because of the risk of neurotoxicity (APS 2016; ISMP 2007).

Metabolism/Transport Effects

Substrate of CYP2B6 (Major), CYP2C19 (Minor), CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acyclovir-Valacyclovir: May increase serum concentrations of Meperidine. Acyclovir-Valacyclovir may increase active metabolite exposure of Meperidine. *Risk C: Monitor*

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Almotriptan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Alosetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amifampridine: Agents With Seizure Threshold Lowering Potential may increase neuroexcitatory and/or seizure-potentiating effects of Amifampridine. *Risk C: Monitor*

Amisulpride (Oral): Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Amisulpride (Oral). Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Amphetamines: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability). Initiate amphetamines at lower doses, monitor frequently, and adjust doses as needed. *Risk C: Monitor*

Antiemetics (5HT3 Antagonists): May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Antipsychotic Agents: Serotonergic Agents (High Risk) may increase adverse/toxic effects of Antipsychotic Agents. Specifically, serotonergic agents may enhance dopamine blockade, possibly increasing the risk for neuroleptic malignant syndrome. Antipsychotic Agents may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

ARIPiprazole Lauroxil: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of ARIPiprazole Lauroxil. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

ARIPiprazole: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of ARIPiprazole. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Benperidol. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Blonanserin: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Blonanserin. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Brexpiprazole: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Brexpiprazole. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Bromperidol. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BuPROPion: May increase neuroexcitatory and/or seizure-potentiating effects of Agents With Seizure Threshold Lowering Potential. *Risk C: Monitor*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

BusPIRone: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Cariprazine: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Cariprazine. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase serum concentrations of Meperidine. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

CYP2B6 Inducers (Moderate): May decrease serum concentrations of Meperidine. CYP2B6 Inducers (Moderate) may increase active metabolite exposure of Meperidine. Specifically, concentrations of normeperidine, the CNS stimulating metabolite, may be increased. *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of Meperidine. CYP3A4 Inducers (Moderate) may increase active metabolite exposure of Meperidine. Specifically, concentrations of normeperidine, the CNS stimulating metabolite, may be increased. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Meperidine. CYP3A4 Inducers (Strong) may increase active metabolite exposure of Meperidine. Specifically, concentrations of normeperidine, the CNS stimulating metabolite, may be increased. *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Meperidine. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Meperidine. *Risk C: Monitor*

Dapoxetine: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Do not use serotonergic agents (high risk) with dapoxetine or within 7 days of serotonergic agent discontinuation. Do not use dapoxetine within 14 days of monoamine oxidase inhibitor use. Dapoxetine labeling lists this combination as contraindicated. *Risk X: Avoid*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Dexmethylphenidate-Methylphenidate: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Dextromethorphan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

Eletriptan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Eluxadoline: Opioid Agonists may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ergot Derivatives: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

FentaNYL: Meperidine may increase serotonergic effects of FentaNYL. This could result in serotonin syndrome. Meperidine may increase CNS depressant effects of FentaNYL. Management: Consider alternatives to this combination. If use is necessary, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Fexinidazole: May decrease serum concentrations of CYP2B6 Substrates (High risk with Inducers). Management: Avoid concomitant use of fexinidazole and CYP2B6 substrates when possible. If combined, monitor for reduced efficacy of the CYP2B6 substrate. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepirone: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of Meperidine. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Iohexol: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Iohexol. Specifically, the risk for seizures may be increased. Management: Discontinue agents that may lower the seizure threshold 48 hours prior to intrathecal use of iohexol. Wait at least 24 hours after the procedure to resume such agents. In nonelective procedures, consider use of prophylactic antiseizure drugs. *Risk D: Consider Therapy Modification*

Iomeprol: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Iomeprol. Specifically, the risk for seizures may be increased. Management: Discontinue agents that may lower the seizure threshold 48 hours prior to intrathecal use of iomeprol. Wait at least 24 hours after the procedure to resume such agents. In nonelective procedures, consider use of prophylactic antiseizure drugs. *Risk D: Consider Therapy Modification*

Iopamidol: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Iopamidol. Specifically, the risk for seizures may be increased. Management: Discontinue agents that may lower the seizure threshold 48 hours prior to intrathecal use of iopamidol. Wait at least 24 hours after the procedure to resume such agents. In nonelective procedures, consider use of prophylactic antiseizure drugs. *Risk D: Consider Therapy Modification*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lindane: Agents With Seizure Threshold Lowering Potential may increase neuroexcitatory and/or seizure-potentiating effects of Lindane. *Risk C: Monitor*

Linezolid: May increase serotonergic effects of Serotonergic Opioids (High Risk). This could result in serotonin syndrome. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lumacaftor and Ivacaftor: May decrease serum concentrations of CYP2B6 Substrates (High risk with Inducers). *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Methylene Blue: May increase serotonergic effects of Serotonergic Opioids (High Risk). This could result in serotonin syndrome. *Risk X: Avoid*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metoclopramide: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Consider monitoring for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

MiFEPRIStone: May increase serum concentrations of CYP2B6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Monoamine Oxidase Inhibitors (Antidepressant): Meperidine may increase serotonergic effects of Monoamine Oxidase Inhibitors (Antidepressant). This could result in serotonin syndrome. *Risk X: Avoid*

Monoamine Oxidase Inhibitors (Type B): Serotonergic Opioids (High Risk) may increase serotonergic effects of Monoamine Oxidase Inhibitors (Type B). This could result in serotonin syndrome. *Risk X: Avoid*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nefazodone: May increase serotonergic effects of Meperidine. This could result in serotonin syndrome. Nefazodone may increase serum concentrations of Meperidine. Management: Consider reducing meperidine dose. Monitor for signs and symptoms of respiratory depression, sedation, and serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia) when these agents are combined. *Risk D: Consider Therapy Modification*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ondansetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid agonists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opipramol: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Oxitriptan: Serotonergic Agents (High Risk) may increase serotonergic effects of Oxitriptan. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Periciazine: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Periciazine. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

PHENobarbital: May increase CNS depressant effects of Meperidine. PHENobarbital may increase active metabolite exposure of Meperidine. Management: Avoid concomitant use of meperidine and phenobarbital when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosages and duration of each drug. *Risk D: Consider Therapy Modification*

Pipamperone: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Pipamperone. Specifically, the risk of seizures may be increased. *Risk X: Avoid*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Polyethylene Glycol-Electrolyte Solution: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Polyethylene Glycol-Electrolyte Solution. Specifically, the risk of seizure may be increased. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Primidone: May increase CNS depressant effects of Meperidine. Primidone may increase active metabolite exposure of Meperidine. Management: Avoid concomitant use of meperidine and primidone when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosages and duration of each drug. *Risk D: Consider Therapy Modification*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Promazine. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Ramosetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Selective Serotonin Reuptake Inhibitor: Serotonergic Opioids (High Risk) may increase serotonergic effects of Selective Serotonin Reuptake Inhibitor. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Serotonergic Agents (High Risk, Miscellaneous): Serotonergic Opioids (High Risk) may increase serotonergic effects of Serotonergic Agents (High Risk, Miscellaneous). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Serotonergic Non-Opioid CNS Depressants: May increase serotonergic effects of Serotonergic Opioids (High Risk). This could result in serotonin syndrome. Serotonergic Non-Opioid CNS Depressants may increase CNS depressant effects of Serotonergic Opioids (High Risk). Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Serotonin 5-HT1D Receptor Agonists (Triptans): May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: Meperidine may increase serotonergic effects of Serotonin/Norepinephrine Reuptake Inhibitor. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Sertindole: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Sertindole. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Sodium Phosphates: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Sodium Phosphates. Specifically, the risk of seizure or loss of consciousness may be increased in patients with significant sodium phosphate-induced fluid or electrolyte abnormalities. *Risk C: Monitor*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

St John's Wort: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. St John's Wort may decrease serum concentrations of Serotonergic Agents (High Risk). Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Sulpiride: Agents With Seizure Threshold Lowering Potential may increase adverse/toxic effects of Sulpiride. Specifically, the risk of seizures may be increased. *Risk C: Monitor*

Sulpiride: Serotonergic Agents (High Risk) may increase adverse/toxic effects of Sulpiride. Specifically, serotonergic agents may enhance dopamine blockade, possibly increasing the risk for neuroleptic malignant syndrome. Sulpiride may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

Syrian Rue: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Tedizolid: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Consider monitoring for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thiotepa: May increase serum concentrations of CYP2B6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraMADol: Serotonergic Opioids (High Risk) may increase serotonergic effects of TraMADol. This could result in serotonin syndrome. Serotonergic Opioids (High Risk) may increase CNS depressant effects of TraMADol. Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Tricyclic Antidepressants: Serotonergic Opioids (High Risk) may increase serotonergic effects of Tricyclic Antidepressants. This could result in serotonin syndrome. Tricyclic Antidepressants may increase CNS depressant effects of Serotonergic Opioids (High Risk). Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Chronic opioid use may cause hypogonadism and hyperprolactinemia, which may decrease fertility in patients of reproductive potential. Menstrual cycle disorders (including amenorrhea), erectile dysfunction, and impotence have been reported. The incidence of hypogonadism may be increased with the use of opioids in high doses or long-acting opioid formulations. It is not known if the effects on fertility are reversible. Monitor patients on long-term therapy (de Vries 2020; Gadelha 2022).

Consider family planning, contraception, and the effects on fertility prior to prescribing opioids for chronic pain to patients who could become pregnant (ACOG 2017; CDC [Dowell 2022]).

Pregnancy Considerations

Meperidine and the active metabolite normeperidine cross the placenta (Malek 2011).

Maternal use of opioids may be associated with poor fetal growth, stillbirth, and preterm delivery (CDC [Dowell 2022]). Opioids used as part of obstetric analgesia/anesthesia during labor and delivery may temporarily affect the fetal heart rate (ACOG 2019).

Neonatal abstinence syndrome (NAS)/neonatal opioid withdrawal syndrome (NOWS) may occur following prolonged in utero exposure to opioids (CDC [Dowell 2022]). NAS/NOWS may be life-threatening if not recognized and treated and requires management according to protocols developed by neonatology experts. Presentation of symptoms varies by opioid characteristics (eg, immediate release, sustained release), time of last dose prior to delivery, drug metabolism (maternal, placental, and infant), net placental transfer, as well as other factors (AAP [Hudak 2012]; AAP [Patrick 2020]). Clinical signs characteristic of withdrawal following in utero opioid exposure include excessive crying or easily irritable, fragmented sleep (<2 to 3 hours after feeding), tremors, increased muscle tone, or GI dysfunction (hyperphagia, poor feeding, feeding intolerance, watery or loose stools) (Jilani 2022). NAS/NOWS occurs following chronic opioid exposure and would not be expected following the use of opioids at delivery (AAP [Patrick 2020]).

Monitor infants of mothers on long-term/chronic opioid therapy for symptoms of withdrawal. Symptom onset reflects the half-life of the opioid used. Monitor infants for at least 3 days following exposure to immediate-release opioids; monitor for at least 4 to 7 days following exposure to sustained-release opioids (AAP [Patrick 2020]; CDC [Dowell 2022]). Monitor newborns for excess sedation and respiratory depression when opioids are used during labor.

When opioids are needed to treat acute pain in pregnant patients, the lowest effective dose for only the expected duration of pain should be prescribed (CDC [Dowell 2022]).

Although approved for use in obstetrical analgesia, meperidine is not recommended for peripartum analgesia due to the prolonged half-life of the active metabolite in the mother and neonate (ACOG 2019). Opioid use for pain following vaginal or cesarean delivery should be made as part of a shared decision-making process. A stepwise multimodal approach to managing postpartum pain is recommended. A low-dose, low-potency, short-acting opioid can be used to treat acute pain associated with delivery when needed (ACOG 2021).

Opioids are not preferred for the treatment of chronic noncancer pain during pregnancy; consider strategies to minimize or avoid opioid use. Advise pregnant patients requiring long-term opioid use of the risk of NAS/NOWS and provide appropriate treatment for the neonate after delivery. NAS/NOWS is an expected and treatable condition following chronic opioid use during pregnancy and should not be the only reason to avoid treating pain with an opioid in pregnant patients (ACOG 2017; CDC [Dowell 2022]). Do not abruptly discontinue opioids during pregnancy; taper prior to discontinuation when appropriate, considering the risks to the pregnant patient and fetus if maternal withdrawal occurs (CDC [Dowell 2022]).

Breastfeeding Considerations

Meperidine is present in breast milk.

Multiple reports summarize data related to the presence of meperidine and normeperidine in breast milk (Al-Tamimi 2011; Borgatta 1997; Freeborn 1980; Quinn 1986; Wittels 1990). Exposure to meperidine and normeperidine via breast milk is consistently associated with neonatal sedation and may interfere with breastfeeding. The half-life of the active metabolite normeperidine is prolonged and accumulation in a breastfed infant may occur (AAP [Sachs 2013]; ABM [Martin 2018]; ABM [Reece-Stremtan 2017]).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Nonopioid analgesics are preferred for lactating patients who require pain control peripartum or for surgery outside of the postpartum period. When opioids are needed for lactating patients, use the lowest effective dose for the shortest duration of time to limit adverse events in the mother and breastfeeding infant. A single, occasional dose of meperidine may be generally compatible with breastfeeding. However, when an opioid is needed to treat maternal pain, meperidine is not preferred (AAP [Sachs 2013]; ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; WHO 2002).

When chronic opioids are prescribed prenatally and continued postpartum, breastfeeding may be initiated to help mitigate potential newborn withdrawal; monitor both the mother and the infant (AAP [Meek 2022]; AAP [Patrick 2020]).

Monitor infants exposed to opioids via breast milk for drowsiness, sedation, feeding difficulties, or limpness (ACOG 2019). Withdrawal symptoms may occur when maternal use is discontinued, or breastfeeding is stopped.

Monitoring Parameters

Pain relief, respiratory and mental status, blood pressure; bowel function; signs/symptoms of misuse, abuse and substance use disorder; signs or symptoms of hypogonadism or hypoadrenalism (Brennan 2013); serotonin syndrome in patient receiving other medications that enhance serotonergic activity (Gillman 2005); signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [noncardiac], dysphagia, regurgitation).

Mechanism of Action

Binds to opioid receptors in the CNS, causing inhibition of ascending pain pathways, altering the perception of and response to pain; produces generalized CNS depression

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Analgesic: Oral, IM, SubQ: 10 to 15 minutes; IV: ~5 minutes

Peak effect: IV: 5 to 7 minutes; IM, SubQ: ~1 hour; Oral: 2 hours

Duration: Oral, IM, SubQ: 2 to 4 hours; IV: 2 to 3 hours

Absorption: IM, Oral: Erratic and highly variable

Distribution: V_{dss}:

Neonates: Preterm 1 to 7 days: 8.8 L/kg; Term 1 to 7 days: 5.6 L/kg

Infants 1 week to 2 months: 8 L/kg

Infants and Children 3 to 18 months: 5 L/kg

Children 5 to 8 years: 2.8 L/kg

Adults: 3 to 4 L/kg

Protein binding (to alpha 1-acid glycoprotein): Neonates: 52%; Infants: 3 to 18 months: 85%; Adults: 65% to 75%

Metabolism: Hepatic; hydrolyzed to meperidinic acid (inactive) or undergoes N-demethylation to normeperidine (active; has $^{1}/_{2}$ the analgesic effect and 2 to 3 times the CNS effects of meperidine)

Bioavailability: ~50% to 60%; increased with liver disease

Half-life elimination:

Parent drug: Terminal phase:

Preterm infants 3.6 to 65 days of age: 11.9 hours (range: 3.3 to 59.4 hours)

Term infants: 0.3 to 4 days of age: 10.7 hours (range: 4.9 to 16.8 hours); 26 to 73 days of age: 8.2 hours (range: 5.7 to 31.7 hours)

Neonates: 23 hours (range: 12 to 39 hours)

Infants 3 to 18 months: 2.3 hours

Children 5 to 8 years: 3 hours

Adults: 2.5 to 4 hours, Liver disease: 7 to 11 hours

Normeperidine (active metabolite): Neonates: 30 to 85 hours; Adults: 8 to 16 hours; normeperidine half-life is dependent on renal function and can accumulate with high doses or in patients with decreased renal function; normeperidine may precipitate tremors or seizures

Excretion: Urine (as metabolites; ~5% as unchanged drug)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Accumulation of meperidine and/or normeperidine may occur.

Hepatic function impairment: Accumulation of meperidine and/or normeperidine may occur. Half-life is 1.3 to 2 times greater in cirrhotic patients.

Older adult: Elderly patients have a slower elimination rate.

Postoperative patients: The half-life is 1.3 to 2 times greater in these patients.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Demerol tablets, oral solution: https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/005010s062lbl.pdf

Patient Counseling Points

(For additional information see ”Meperidine (pethidine): Patient drug information”)

What is this drug used for?

• It is used to manage pain when non-opioid pain drugs do not treat your pain well enough or you cannot take them.

• It is used during surgery.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Constipation

• Feeling dizzy, sleepy, tired, or weak

• Flushing

• Upset stomach or throwing up

• Sweating a lot

• Headache

• Stomach pain

• Dry mouth

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating

• Severe dizziness or passing out

• Feeling confused

• Seizures

• Chest pain or pressure

• Fast, slow, or abnormal heartbeat

• Trouble passing urine

• Hallucinations (seeing or hearing things that are not there)

• Mood changes

• Severe bowel problem like severe constipation or stomach pain

• Trouble breathing, slow breathing, or shallow breathing

• Noisy breathing

• Breathing problems during sleep (sleep apnea)

• Shakiness

• Trouble controlling body movements

• Change in eyesight

• Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache

• Severe adrenal gland problem like if you feel very tired or weak, you pass out, or you have severe dizziness, very upset stomach, throwing up, or decreased appetite

• Lowered interest in sex, fertility problems, no menstrual period, or ejaculation problems

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Meperidina chobet;
- (AU) Australia: Pethidine HCL;
- (BE) Belgium: Dolantine;
- (BR) Brazil: Dolantina | Petinan;
- (CH) Switzerland: Pethidin streuli;
- (CL) Chile: Demerol;
- (CN) China: Pethidine;
- (CZ) Czech Republic: Dolsin;
- (DE) Germany: Dolantin;
- (EE) Estonia: Petidin dak;
- (ET) Ethiopia: Pethidine;
- (FI) Finland: Petidin;
- (GB) United Kingdom: Pethidine | Pethidine roche;
- (GR) Greece: Pethidine;
- (HK) Hong Kong: Pethidine;
- (HU) Hungary: Dolargan;
- (IE) Ireland: Pethidine | Pethilan;
- (IL) Israel: Dolestine;
- (IT) Italy: Meperidina;
- (KE) Kenya: Pethidine;

- (LT) Lithuania: Dolsin;
- (LU) Luxembourg: Dolantine;
- (LV) Latvia: Dolsin;
- (NL) Netherlands: Pethidini hcl pch;
- (PL) Poland: Dolargan | Dolsin;
- (PR) Puerto Rico: Demerol | Meperidine HCL | Meperidine hydrochloride | Meperitab;
- (SG) Singapore: Pethidine;
- (SK) Slovakia: Dolsin;
- (TH) Thailand: Pangesic;
- (TR) Turkey: Aldolan;
- (ZA) South Africa: Pethidine

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